

NAVEED AHMED

Principal Software Engineer | Enterprise & AI Systems Architect

Karachi, Sindh, Pakistan (*Open & Planning to Relocate to Auckland, New Zealand*)
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EXECUTIVE SUMMARY

Senior / Principal Software Engineer with 9+ years of hands-on experience designing and delivering enterprise-grade distributed systems, cloud microservices, and modern polyglot architectures across international clients. Proven track record in core engineering leadership: defining technical strategy alongside CTOs, taking research proofs-of-concept (PoCs) to high-throughput production, modernizing legacy systems, and establishing standardized CI/CD and deployment architectures across Azure and AWS. Pioneer in embedding LLM-assisted workflows, Model Context Protocol (MCP) tooling, and multi-agent systems into enterprise software development lifecycles.

CORE COMPETENCIES & TECHNICAL SKILLS

- Enterprise & Systems Languages:** C# / .NET (6/8), TypeScript, JavaScript, Python, C++, Rust (foundations & systems tooling), Java.
- Frontend & Web Engineering:** TypeScript, React.js, Next.js, State Management, Modern Web Performance, Responsive Component Design.
- Distributed Systems & Architecture:** Microservices, Event-Driven Architectures, Domain-Driven Design (DDD), Asynchronous Processing, High-Throughput Messaging (**Apache Kafka**, **Azure Service Bus**, **AWS EventBridge**, **Hangfire**).
- Cloud & DevOps / IaC:** Azure (AKS, App Services, Blob, Cosmos DB, Service Bus), AWS, **Docker**, **Kubernetes**, **Terraform** / **IaC**, CI/CD Pipelines (GitHub Actions, Azure DevOps).
- AI-Assisted Development & LLM Engineering:** LLM-Assisted Engineering Workflows, **Model Context Protocol (MCP)** tool development, Multi-Agent Systems (**LangGraph**), RAG Architectures, Vector Databases (Pinecone), Guardrails & Evaluation.
- Engineering Leadership:** Technical Strategy, CTO/Stakeholder Advisory, Architecture Blueprinting, PoC-to-Production Delivery, Mentoring & Code Reviews.

PROFESSIONAL EXPERIENCE

Principal Software Engineer | 10Pearls

Jan 2025 – Present | Karachi, PK (*International Clients*)

- Core Technical Strategy:** Partner with executive leadership and client CTOs to define technical roadmaps, architectural standards, and engineering strategies for large-scale enterprise platforms.
- PoC to Production Engineering:** Spearheaded the transition of advanced AI and distributed computing proofs-of-concept into hardened, enterprise-ready microservices and production endpoints.
- Standardized Tooling & AI-Assisted Workflows:** Built reusable developer tooling, including Model Context Protocol (MCP) sidecars and automated code-generation harnesses, reducing delivery cycle times across polyglot service teams.
- Architecture & Infrastructure:** Architected cloud-native services across Azure and AWS with containerized orchestration (Docker/AKS), automated CI/CD pipelines, and secure enterprise integration patterns.
- Mentorship:** Mentored senior and mid-level engineers on modern architectural patterns, distributed system observability, and LLM-assisted development practices.

Technical Lead | Turing.com (Contract)

Jul 2022 – Jan 2025 | Remote, US

- Greenfield Enterprise SaaS Delivery:** Owned end-to-end architecture and implementation for a US healthcare facility management SaaS platform, migrating legacy desktop systems to a modern, scalable cloud microservices architecture.
- Domain-Driven Design (DDD):** Decomposed complex monolithic business logic into bounded domain services (scheduling, shift allocation, payroll, time-tracking) with asynchronous background processing via Hangfire.
- Polyglot & Cloud Infrastructure:** Delivered robust REST/TypeScript APIs, React frontends, and backend microservices deployed on Azure (AKS, App Service, Azure SQL, Azure AD SSO).
- Engineering Standards & Mentorship:** Led and coached a distributed team of engineers, instituting automated testing standards, CI/CD pipelines, and strict code review gates.

Senior Software Engineer | VentureDive

Sep 2020 – Apr 2022 | Remote

- High-Throughput Distributed Systems:** Built and scaled financial platforms, payment gateway integrations, and multi-tenant SaaS applications for international clients (Daraz, Motiv, PayActiv).
- Event-Driven Messaging:** Engineered real-time data pipelines and messaging integrations utilizing Apache Kafka, Azure Service Bus, and AWS EventBridge to handle high transactional volume with fault tolerance.
- Performance Optimization:** Conducted deep technical audits and profiling across .NET, Node.js, and PostgreSQL services, reducing latency and scaling operational capacity.

Senior Software Engineer I Stingray Technologies (Pvt) Ltd

Oct 2017 – Aug 2020 | Karachi, PK

- **Mission-Critical Systems:** Developed and maintained mission-critical software systems and real-time simulators for defense equipment under strict low-latency and high-reliability constraints.
- **Low-Level & Distributed Messaging:** Designed core architectures handling thousands of real-time messages per second using C++, C#, multi-threaded concurrency models, and distributed databases (MongoDB, SQL).
- **Hardware-Software Integration:** Built hardware simulators to validate and exercise equipment communication protocols prior to field deployment.

Software Engineer I Techlogix

Dec 2016 – Oct 2017 | Karachi, PK

- **Enterprise Web Services:** Developed scalable backend services and web APIs using C#, .NET Web API, and SQL Server for enterprise clients.

FEATURED ENGINEERING PROJECTS

1. Conversational Database Interface & Developer Tooling (MCP)

Python, TypeScript, React, Model Context Protocol (MCP), LLMs

- Developed custom MCP sidecars and server tools that enable LLM agents to safely interact with live enterprise database schemas, translate natural language into optimized queries, and self-validate outputs with explainability guardrails.

2. Autonomous Multi-Agent Orchestration Platform

Python, TypeScript, React, Docker, PostgreSQL, LangGraph

- Engineered a cooperative multi-agent system executing parallel tasks (content generation, strategy, and optimization) integrated with external APIs, complete with human-in-the-loop verification and brand-safety guardrails.

3. Enterprise Document Intelligence & High-Fidelity RAG Engine

Python, React, Pinecone Vector DB, OpenAI Embedding APIs, Azure

- Re-engineered a legacy RAG platform encountering critical processing bottlenecks; designed custom citation-mapping algorithms linking generated synthesis back to exact document clauses and page coordinates for compliance auditing.

4. Healthcare Workforce Scheduling & Distributed Backend

C# / .NET, TypeScript, React, Azure Kubernetes Service (AKS), Hangfire, Azure SQL

- Modernized desktop facility management rules into scalable domain services on AKS, supporting thousands of concurrent healthcare shift records and real-time roster updates.

EDUCATION

Bachelor of Engineering (Computer Engineering)

NED University of Engineering and Technology | 2013 – 2016 (CGPA: 3.86 / 4.00)